Younes Mouhib

MSc Mathematics, ETH Zurich • data science · optimisation · quantitative analysis



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Nationality French
Work permit B permit
Availability immediately

PROFILE

Seeking a job or internship in data science, optimisation or quantitative analysis. Serious, reliable, motivated and responsible: a fast learner who works independently and builds projects end to end. Sharp critical thinker, curious, with broad interests.

SKILLS

Programming Python · SQL · MATLAB · OCaml · Lean 4
Libraries NumPy · pandas · scikit-learn · PyTorch
Optimisation Gurobi · SAT solvers (PySAT)
Tools Git · CI · Docker · Linux · LaTeX · Excel
Languages French C2 · English C1 · Arabic C1 · Spanish B2 · Indonesian B1 · German A1

EDUCATION

09.2023 – 08.2026 **ETH Zurich — MSc Mathematics** · final grade 5.21/6 — Zurich
Network & Integer Optimization · Convex Optimization and Applications · Structural Graph Theory · Graph Algorithms · Algebraic Methods in Combinatorics · Algebraic Topology · Algebraic Geometry

09.2018 – 07.2023 **EPFL — BSc Mathematics** — Lausanne
Probability · Measure & Integration · Ergodic Theory · Stochastic Processes · Statistics · Time Series · Numerical Analysis · Discrete Optimization

2016 – 2018 **Lycée Masséna — Classes préparatoires MPSI/MP*** — Nice
Selective two-year programme in mathematics, physics & computer science • top 10% of class

RESEARCH

2026 **Master's thesis, ETH Zurich (Prof. Dr. Lorenz Halbeisen)** · grade 6.0/6 — GitHub
SAT-based search: a symmetry reduction turned an intractable problem into a small constraint model, setting new record bounds for the Hales–Jewett numbers, independently verified by code.

2026 **Preprint (sole author)**: "Lower Bounds for the Hales–Jewett Numbers via Symmetric and One-Weight Colorings", arXiv: 2606.22155

2023 **Bachelor thesis, EPFL (Prof. Dr. Maria Colombo, Dr. Lucio Galeati)**
"On the Subadditive Ergodic Theorem and Its Applications"

PROJECTS

Data science **Football injury risk** · weekly model and live match-day tool — GitHub
Calibrated logistic regression on 1.29M player-weeks: on a held-out season, the top-risk 10% were injured six times as often as the lowest 10%. Live tool: offline app for the bench, validated on synthetic matches.

Quant research **Crypto momentum strategy (Binance)** — GitHub
Cost-aware backtesting with walk-forward validation, reporting out-of-sample results only, with bootstrap confidence intervals and a deflated Sharpe ratio; 137 tests, CI on every push.

Machine learning **Sudoku vision** — GitHub
A CNN (PyTorch) reads handwritten sudoku grids, a SAT solver solves them and MaxSAT repairs misread digits: fully correct grids rise from 75% to 99% on 1,000 test puzzles.

AI agents **Brigade, multi-agent LLM prover** — GitHub
LLM agents plan and write mathematical proofs; only the Lean 4 proof checker can accept them. FastAPI backend, Docker image, CI.

TEACHING & PROFESSIONAL EXPERIENCE

06.2026 **Course Instructor (PVK), Topology — ETH Zurich**: 4-day exam-preparation course, 20+ students

2018 – 2026 **Part-time work alongside full-time study (studies self-financed throughout)**
FMEL Lausanne (warehouse & logistics), Tribeca and Le Cailhau (waiter), Uber (delivery courier), McDonald's Zurich (crew member), private tuition • payslips on request

07.2021 – 01.2023 **Homework Tutor, maths and science (grades 9–11) — Collèges Villamont and St-Roch**

04.2022 **Science Outreach Volunteer — Science Day in Thun, EPFL**: hands-on science activities

09.2020 – 02.2021 **Student Mentor — EPFL**: mentored first-year students during the COVID-19 pandemic

References: contact details available on request.